APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

How a Global Bank Runs: A Quantitative Operating Model

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Regulatory Framework: Educational synthesis referencing Basel-style market, credit, liquidity, and model-risk frameworks

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1 Purpose and Scope

This document explains how a large global bank operates from a quantitative perspective. The goal is educational: to make explicit the mathematics, data flows, compute requirements, and organizational conflicts hidden behind trading revenue, risk reports, treasury charges, and regulatory capital numbers.

The emphasis is not on one single model. A bank is a coupled system of desks, legal entities, balance-sheet constraints, pricing services, risk engines, accounting rules, collateral agreements, treasury transfer pricing, and governance. The same trade is simultaneously:

- a client relationship event for the front office,
- a market-risk vector for risk management,
- a future exposure profile for credit and XVA,
- a funding and liquidity burden for treasury,
- a reserve, capital, and ledger object for finance,
- and a validation and audit subject for second and third line control functions.

This is why the bank should be viewed as a constrained optimization problem rather than a collection of isolated desks.

2 The Bank as a Constrained Optimization Problem

2.1 Economic Objective

At a high level, management wants to maximize long-run risk-adjusted franchise value, not raw trading revenue:

$$\max_x \ \Pi(x) = \mathbb{E}[\text{Revenue}(x)] - \text{EL}(x) - \text{FundingCost}(x) - \text{LiquidityCost}(x) - \text{OperatingCost}(x) - \text{CapitalCharge}(x)$$

subject to market, credit, liquidity, leverage, conduct, and legal-entity constraints.

In practice, the bank does not solve one global optimization exactly. Instead, it decomposes the problem across desks and functions, then re-couples them through prices, limits, transfer charges, and management committees.

2.2 Balance-Sheet Identity

Every bank-level quantitative discussion starts from the identity

$$\text{Assets} = \text{Liabilities} + \text{Equity}.$$

The front office often focuses on marginal trade economics. Treasury and finance care about how that marginal trade changes the entire right-hand side of the identity: funding mix, liquidity buffer, retained earnings, and capital ratios. This difference in viewpoint drives many of the internal conflicts discussed later.

2.3 Business Decomposition

For a universal bank, the main quant-relevant components are:

- **Trading and market making:** equities, rates, FX, credit, commodities, securitized products, and derivatives.
- **Banking book:** loans, deposits, securities portfolios, mortgage and consumer books.
- **Treasury and ALM:** funding, FTP, liquidity buffer management, structural interest-rate risk.
- **Credit risk:** loan default risk, counterparty risk, concentration, underwriting limits.
- **XVA and collateral:** derivative exposure, margining, funding valuation, initial margin cost.
- **Finance and capital:** P&L explain, reserves, accounting classification, regulatory capital, performance measurement.
- **Model risk and validation:** independent review, challenger models, backtesting, governance.

3 Front Office Trading and Pricing

3.1 What Trading Desks Actually Do

The front office sells liquidity and balance-sheet capacity to clients. A client trade is rarely priced as a pure textbook fair value. The executable quote includes compensation for inventory risk, hedging slippage, counterparty quality, funding consumption, collateral terms, and scarce capital.

3.2 Clean Price, Executable Price, and All-in Price

For a derivative or financing trade, a useful decomposition of the *client executable price* is

$$P_{\text{client}} = P_{\text{clean}} + \text{CVA} + \text{DVA} + \text{FVA} + \text{MVA} + \text{KVA} + \text{LiquidityPremium} + \text{BidAsk}.$$

Interpretation:

- P_{clean} is the risk-free or collateral-consistent theoretical value.
- CVA charges for counterparty default risk.
- DVA reflects the bank's own default risk in valuation terms; it is usually treated carefully in management reporting because it is not equivalent to cash earnings.
- FVA charges for unsecured or partially secured funding.
- MVA charges for the cost of funding segregated initial margin.
- KVA allocates the economic cost of capital tied up by the trade.

This equation should **not** be read as a universal accounting fair-value identity. In many banks, KVA, liquidity premium, and bid-ask are internal pricing or execution add-ons rather than pure accounting valuation adjustments, while sign conventions for DVA and FVA vary by policy and product set. The stable point is narrower: the executable trade price is a balance-sheet price, not just a clean model price.

3.3 Daily P&L Mechanics

For a portfolio value $V(S, \sigma, t)$ driven by market factors S and volatility state σ, daily P&L explain is approximated by

$$\Delta\text{P\&L} \approx \sum_i \Delta_i \Delta S_i + \frac{1}{2}\sum_{i,j} \Gamma_{ij} \Delta S_i \Delta S_j + \sum_k \text{Vega}_k \Delta\sigma_k + \Theta\Delta t + \text{Carry} + \text{NewTrade} + \text{Fees} - \text{HedgeCost}.$$

For simple linear products, this approximation is often enough. For structured products and large option books, the bank uses full revaluation under shocked curves, surfaces, and correlations.

3.4 Pre-trade Decision Rule

A practical desk-level optimization is

$$\begin{aligned} &\max_q \ \text{ClientSpread}(q) - \text{ExpectedHedgeCost}(q) - \text{XVA}(q) - \text{FTP}(q) - \text{CapitalCharge}(q) \\ &\text{s.t.} \quad \text{LimitUtilization}(q) \le 100\%, \quad \text{CCRHeadroom}(q) \ge 0, \quad \text{LiquidityConsumption}(q) \le \text{budget}. \end{aligned}$$

This is where the apparent conflict between "sales wants to win the trade" and "risk wants to constrain the trade" becomes mathematically explicit.

4 Market Risk

4.1 Sensitivities and Local Approximation

Let S be the vector of market risk factors. The standard second-order local approximation is

$$\Delta V \approx \nabla_S V \cdot \Delta S + \frac{1}{2}\Delta S^\top \nabla_S^2 V \Delta S.$$

This approximation is useful for intraday risk attribution, but it is not exact for strongly nonlinear books, callable structures, path-dependent exotics, or products sensitive to surface dynamics and jumps.

4.2 VaR and Expected Shortfall

For a P&L distribution generated by scenarios $\{\text{P\&L}^{(n)}\}_{n=1}^N$,

$$\text{VaR}_\alpha = -Q_\alpha(\text{P\&L}), \qquad \text{ES}_\alpha = -\mathbb{E}[\text{P\&L} \mid \text{P\&L} \le Q_\alpha(\text{P\&L})].$$

For a linear-Gaussian approximation with exposure vector w and factor covariance matrix Σ,

$$\mathrm{VaR}_\alpha^{\mathrm{param}} = z_\alpha \sqrt{w^\top \Sigma w}.$$

This is fast and useful for limit checks, but it misses skew, kurtosis, optionality, and jump-to-default effects. Large banks therefore operate several market-risk views simultaneously:

- fast parametric approximations for pre-trade checks,
- historical and filtered historical simulation for core limit reporting,
- expected shortfall style tail metrics for capital and governance,
- stress scenarios and reverse stress for non-local losses.

For modern trading-book capital, the important distinction is that expected shortfall with liquidity-horizon style adjustments is the regulatory tail metric in FRTB-style internal-model frameworks, while VaR remains more common for internal limits, legacy views, and fast approximations.

4.3 Aggregation Problem

Risk is measured at multiple levels:

- trade,
- trader,
- desk,
- business line,
- legal entity,
- and group.

Aggregation is not trivial because diversification, concentration, liquidity horizons, and legal-entity restrictions all matter. The key failure mode is believing firm-wide diversification is fully usable when the actual constraint binds at desk or entity level.

4.4 Stress Testing

If $x^{(0)}$ is the current market state and $x^{(s)}$ a stressed state, scenario loss is

$$L^{(s)} = V(x^{(0)}) - V(x^{(s)}).$$

Useful scenarios are not just historical replays. A bank typically uses:

- **historical scenarios** such as crisis snapshots,
- **hypothetical scenarios** such as sovereign spread blowouts,
- **reverse stress** to find the smallest shock set that breaks a limit or ratio.

The organizational importance is that stress outputs propagate beyond market risk into liquidity, collateral, and capital planning.

5 Counterparty Credit Risk, XVA, and Collateral

5.1 Exposure Profiles

For a future netting-set value $V(t)$, positive exposure is

$$E(t) = \max(V(t), 0).$$

Expected exposure and expected positive exposure are then

$$\mathrm{EE}(t) = \mathbb{E}[E(t)], \qquad \mathrm{EPE} = \frac{1}{T}\int_0^T \mathrm{EE}(t)\,dt$$

and potential future exposure at confidence level α is

$$\mathrm{PFE}_\alpha(t) = Q_\alpha(E(t)).$$

These objects drive limit monitoring, collateral calls, initial margin calibration, and pricing adjustments.

5.2 CVA and Related Adjustments

At global-bank scale, counterparty risk is managed primarily at the *netting-set* level, not trade by trade. The exposure object used for pricing and limits is therefore typically post-netting and, where relevant, post-collateral with explicit margin-period-of-risk treatment.

The unilateral CVA of a counterparty netting set is commonly written as

$$\mathrm{CVA} = (1 - R_c)\int_0^T \mathrm{EE}^{\mathrm{net,coll}}(t)\,d\mathrm{PD}_c(t)\,D(t),$$

$$\mathrm{CVA} \approx \sum_{i=1}^{m}(1 - R_c)\,\mathrm{EE}^{\mathrm{net,coll}}(t_i)\,\Delta\mathrm{PD}_c(t_i)\,D(t_i).$$

Here R_c is recovery, PD_c is the counterparty default curve, and $D(t)$ is the discount factor. Related adjustments include:

- **DVA:** own-default adjustment,
- **FVA:** funding cost of uncollateralized exposure,
- **MVA:** funding cost of initial margin,
- **ColVA:** collateral rate mismatch effects,
- **KVA:** cost of capital required to support the trade.

5.3 Collateralized Exposure

Under a CSA with variation margin $VM(t)$, threshold TH, and remargining frictions, a simple approximation is

$$E^{\text{net,coll}}(t) = \max\big(V(t) - VM(t) - TH, 0\big) + \text{MPOR AddOn}.$$

Here $V(t)$ should be interpreted as the *netting-set* mark after legally enforceable netting, not as a standalone trade mark. The margin-period-of-risk add-on captures the fact that exposure can regrow between the last successful margin exchange and closeout.

5.4 Wrong-Way Risk

Independent exposure and default are often too optimistic. If exposure tends to rise exactly when counterparty quality deteriorates, then

$$\text{CVA}_{\text{WWR}} > \text{CVA}_{\text{independent}}.$$

Examples include FX trades with emerging-market sovereigns and commodity exposures to producers whose credit quality co-moves with the underlying commodity.

6 Loan Credit Risk and Provisioning

6.1 Expected Loss and Unexpected Loss

For a loan or obligor-level exposure,

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}.$$

Expected loss is a mean quantity. Unexpected loss and tail loss drive capital, portfolio concentration limits, and stress testing.

6.2 IFRS 9 Style Provisioning

An educational stylization of lifetime expected credit loss is

$$\text{ECL} = \sum_{t=1}^{T} \text{PD}_t \times \text{LGD}_t \times \text{EAD}_t \times D(t).$$

Stage migration matters:

- **Stage 1:** typically 12-month loss recognition,
- **Stage 2:** lifetime loss recognition after significant deterioration,
- **Stage 3:** credit-impaired assets.

This means a bank can take a large provisioning hit before default has occurred. That is one reason finance, credit, and the businesses can disagree strongly during deteriorating macro conditions.

6.3 Concentration

Portfolio loss is rarely the sum of independent obligor losses. Sector, geography, sponsor, and wrong-way concentrations create convexity in bad states. A desk may look diversified by name count but still be concentrated in one macro factor.

7 Treasury, ALM, and Internal Funding

7.1 Why Treasury Matters Quantitatively

Treasury is the internal market for balance sheet. Without treasury transfer pricing, desks would systematically overstate profitability by ignoring the cost of long-dated funding, liquidity optionality, collateral drag, and structural rate risk.

7.2 Repricing Gap and NII

For bucket b, repricing gap is

$$\mathrm{Gap}_b = A_b - L_b,$$
$$\Delta\mathrm{NII} \approx \sum_b \mathrm{Gap}_b \Delta r_b.$$

This is a first-order approximation to one-year net interest income sensitivity. It is useful but incomplete because deposits, mortgages, and prepayable assets have behavioral optionality rather than purely contractual maturity.

7.3 Economic Value of Equity

Let $CF(t)$ be net balance-sheet cashflows. Then

$$\mathrm{EVE} = \sum_t D(t)CF(t).$$

Under a duration-gap approximation,

$$\Delta\mathrm{EVE} \approx -DG \times A \times \Delta y,$$
$$DG = D_A - \frac{L}{A} D_L,$$

where D_A and D_L are asset and liability durations. This is still a simplification; embedded options and deposit betas can dominate actual outcomes.

7.4 Funds Transfer Pricing

A stylized internal FTP curve for desk i at tenor T is

$$\mathrm{FTP}_i(T) = r^{\mathrm{base}}(T) + \lambda^{\mathrm{liq}}(T) + c_i^{\mathrm{optional}}(T) + c_i^{\mathrm{capital}}(T) + c_i^{\mathrm{scarcity}}(T).$$

This formula makes internal economics explicit:

- long-dated assets should not be funded as if they were overnight,
- structurally illiquid products should consume a liquidity premium,
- businesses that consume scarce leverage or capital should see a higher internal hurdle rate.

7.5 Liquidity Ratios

Two stylized regulatory constraints are

$$\mathrm{LCR} = \frac{\mathrm{HQLA}}{\text{Net Cash Outflows}_{30d}}, \qquad \mathrm{NSFR} = \frac{\text{Available Stable Funding}}{\text{Required Stable Funding}}.$$

The key practical point is that liquidity can bind before solvency does. A bank can be economically solvent and still fail if it cannot meet withdrawals, roll wholesale funding, or post collateral fast enough.

8 Capital, Performance Measurement, and Limits

8.1 Regulatory and Economic Capital

Risk managers, finance, and treasury distinguish at least four notions:

- accounting reserves,
- regulatory capital,
- economic capital,
- management buffers above binding minima.

Confusing these quantities is a standard source of poor decision-making.

8.2 Core Ratios

The most visible solvency metric is

$$\text{CET1 Ratio} = \frac{\mathrm{CET1}}{\mathrm{RWA}}.$$

A leverage-style metric is

$$\text{Leverage Ratio} = \frac{\text{Tier 1 Capital}}{\text{Exposure Measure}}.$$

Even when a desk believes a trade is low risk, treasury or finance may object because it expands the leverage exposure measure or consumes too much liquidity.

8.3 RAROC

A simplified economic performance metric is

$$\text{RAROC} = \frac{\text{Revenue} - \text{ExpectedLoss} - \text{FundingCost} - \text{OperatingCost}}{\text{EconomicCapital}}.$$

RAROC is not a law of nature, but it is a useful language for adjudicating internal trade-offs. High nominal P&L with poor capital efficiency is often a bad business.

9 Compute, Data, and Systems at Global-Bank Scale

9.1 Why Compute Is a First-Class Constraint

The bank's quantitative workload has two contradictory requirements:

- **low latency** for pricing, quoting, and pre-trade checks,
- **high throughput** for overnight risk, XVA, stress, finance, and regulatory production.

9.2 Event Flow

The minimal event sequence is:

1. market data update or client trade arrives,
2. trade capture normalizes terms and books the event,
3. pricing service marks the trade and computes sensitivities,
4. risk service aggregates Greeks, VaR/ES, scenarios, and breaches,
5. credit/XVA service updates exposures and valuation adjustments,
6. collateral system updates margin and dispute state,
7. treasury re-prices funding and balance-sheet usage,
8. finance consumes the same state for P&L, reserves, and capital reporting.

The dependency graph is the real system. The mathematics is only one layer.

9.3 Order-of-Magnitude Compute

Workload	Typical Scale	Notes
Vanilla pricing request	microseconds to milliseconds	used in quoting and intraday hedging
Structured trade pricing	milliseconds to seconds	tree, PDE, local/stochastic vol, or nested MC approximations
Desk market-risk run	10^4–10^6 revaluations	depending on scenario count and product mix
Bank-wide XVA run	10^7–10^9 trade-path-date evaluations	reduced by netting, regression, and caching
IFRS 9 / retail credit run	millions of accounts over multiple macro scenarios	heavily parallel batch workload
Enterprise stress testing	hundreds to thousands of scenarios across books and horizons	often hybrid full revaluation plus sensitivities

9.4 Complexity Sketches

If a bank has N_{trades} trades, N_{scen} scenarios, and N_{dates} future dates, then brute-force XVA-style exposure work is roughly

$$O(N_{\text{trades}} N_{\text{scen}} N_{\text{dates}}).$$

If a retail credit engine has N_{acct} accounts, N_{macro} scenarios, and N_{qtr} forecast quarters, then a simple workload proxy is

$$O(N_{\text{acct}} N_{\text{macro}} N_{\text{qtr}}).$$

That is why banks rely on:

- distributed CPU grids and, increasingly, GPU acceleration for specific workloads,
- cached curves, vol surfaces, and scenario blocks,
- netting-set aggregation instead of trade-by-trade recomputation where valid,
- surrogate models, regressions, and sensitivity-based approximations for non-material positions,
- strict model versioning and reproducible data snapshots.

9.5 What Fails in Practice

The hard parts are usually:

- stale or inconsistent market data,
- legal-entity mapping errors,
- disagreement between front-office and risk marks,
- incomplete product representation in control systems,

- and non-reproducible overnight runs caused by dependency drift.

10 How the Components Interact and Compete

10.1 Front Office vs Market Risk

The desk wants limit headroom, higher inventory tolerance, and wider ability to warehouse basis risk. Market risk wants conservative aggregation, robust tail metrics, and faster escalation when positions become nonlinear or concentrated.

10.2 Front Office vs XVA

Sales and trading want tighter client prices. XVA wants compensation for counterparty quality, collateral asymmetry, funding tenor, and initial margin. This is not a philosophical disagreement; it is a direct dispute over whether desk-reported revenue is real after balance-sheet costs.

10.3 Front Office vs Treasury

Desks prefer cheap, long-dated internal funding and flexible balance-sheet allocation. Treasury wants stable funding, low liquidity burn, and efficient use of scarce leverage exposure. A repo desk, prime services business, and long-dated rates desk can all be profitable in isolation while competing for the same scarce balance sheet.

10.4 Credit Risk vs Markets

Credit officers focus on default, concentration, and wrong-way risk. Markets focus on spread capture and client flow. A trade can look attractive on expected spread yet be rejected because it fills a concentration bucket or creates tail exposure under a stressed collateral dispute.

10.5 Finance vs Trading

Finance needs explainable, auditable numbers and reserve frameworks that reconcile with the general ledger. Trading wants risk-sensitive fair values and timely P&L. Disputes often arise over valuation reserves, day-one P&L, funding transfer pricing, and whether certain gains are distributable versus purely accounting.

10.6 Model Validation vs Everyone

Model validation is structurally adversarial in a healthy way. Validators are supposed to ask whether the desk's favored model is biased, whether risk approximations understate tails, whether backtesting supports the claimed use, and whether a simpler challenger model reveals fragility. This can delay releases, but it prevents the bank from optimizing around a broken proxy.

10.7 A Useful Internal Value Equation

A stylized desk-level value equation is

$$\text{DeskValue} \approx \text{ClientRevenue} - \text{HedgeCost} - \text{XVA} - \text{FTP} - \text{CapitalCharge} - \text{ExpectedLoss} - \text{OpEx}.$$

The purpose of the control functions is not to eliminate disagreement. It is to force the disagreement to become measurable before it becomes a real loss.

11 An End-to-End Example

Consider a long-dated interest-rate swap with a corporate client.

1. The rates desk computes a clean price from projected floating cashflows and discount factors.
2. Credit and XVA estimate expected exposure, CVA, and the cost of initial margin under the applicable netting set and CSA.
3. Treasury computes the FTP charge implied by tenor, currency, and balance-sheet usage.
4. Market risk updates DV01, curve-bucket sensitivities, stress losses, and desk limit utilization.
5. Credit risk checks counterparty headroom and concentration.
6. Finance determines booking, reserves, and the path by which the trade enters reported P&L and capital.

The quote that finally goes to the client reflects all of these layers. This is why "the trader priced the trade" is an incomplete description of what actually happened.

12 Model-Risk and Educational Caveats

Several caveats are essential for correctness:

- no single model is valid across all products and horizons,
- pricing values, management P&L, accounting values, and regulatory capital numbers are related but not identical,
- simplified formulas such as duration-gap, parametric VaR, or independent-exposure CVA are approximations rather than exact truth,
- diversification is never fully free because legal entities, liquidity horizons, and concentration limits restrict its usability,
- and compute scale claims depend strongly on product mix, booking architecture, scenario design, and approximation policy.

13 Independent Review Outcome

An independent reviewer was asked to validate the document for technical correctness, with focus on the main failure modes in a bank-wide quant explanation: conflating pricing with accounting, conflating expected loss with capital, overstating diversification, understating collateral and liquidity frictions, and presenting approximate methods as exact.

The review identified three material issues in the first draft and all three were corrected in this version:

- the executable pricing formula now explicitly distinguishes accounting fair value from internal charges such as KVA and bid-ask,

- the CCR/XVA section now states that exposure is netting-set and collateral aware, with margin-period-of-risk treatment,
- the market-risk section now clarifies that expected shortfall, not plain VaR, is the key regulatory tail metric in FRTB-style trading-book capital.

Residual caveat: many formulas here are intentionally stylized for teaching and would need product-specific and jurisdiction-specific refinement before production use.

14 Conclusion

From a quant perspective, a global bank is a control system built on top of stochastic valuation, scenario aggregation, funding allocation, capital constraints, and governance. Trading creates revenue opportunities, but risk, treasury, credit, finance, and validation determine whether that revenue is real, fundable, survivable, and reportable.

That is the curtain behind the industry: banks are not only trading machines. They are optimization machines operating under uncertainty, with capital and liquidity as scarce state variables and with mathematics serving as the language that allows competing functions to negotiate over the same balance sheet.